Task-Conditioned Reinforcement Learning (TC-RL)

Definition 1 *Given a Markov Decision Process (MDP) $\mathcal{M} = \langle S, A, P, \iota \rangle$ and a set of tasks Ψ with a prior distribution $\iota_\Psi \in \Delta(\Psi)$, a **task-conditioned policy** is a mapping $\pi : S \times \Psi \to \Delta(A)$. The **Task-Conditioned Reinforcement Learning** (TC-RL) problem is to find a policy π maximizing the probability of satisfying a task $\psi \sim \iota_\Psi$ by navigating the underlying environment $\mathcal{M}$, i.e.,*

$$J(\pi) \triangleq \mathbb{P}_{\substack{\psi \sim \iota_\Psi \\ \tau \sim \mathcal{M}, \pi, \psi}} \left[\tau \models \psi \right], \tag{1}$$

where τ is a trace generated by conditioning π on ψ and running it in $\mathcal{M}$ and $\tau \models \psi$ denotes that the generated behavior τ satisfies the given task ψ. The objective of TC-RL is to solve for the optimal policy $\pi^* \in \arg\max_\pi J(\pi)$, maximizing the probability of satisfying $\psi \sim \iota_\Psi$.

- **Automata-conditioned RL → condition on**

- Instruction-conditioned RL → condition on

RAD Embeddings

Qwen3-Embedding

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condition on

- **Automata-conditioned RL**

- Instruction-conditioned RL

condition on

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RAD Embeddings

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